EduTrace-ES Technology Stack Report

1. Introduction

EduTrace-ES is an Academic Accreditation Status Tracing Platform designed to help educational institutions monitor their accreditation readiness.

The platform integrates embedded systems, environmental sensors, artificial intelligence, document analysis, and a web-based analytics dashboard.

The technology stack for EduTrace-ES combines Internet of Things (IoT), Artificial Intelligence (AI), Large Language Models (LLM),

cloud computing, and modern web technologies to create an intelligent monitoring and accreditation analysis system.

2. Embedded Systems Layer

The embedded layer collects environmental and infrastructure data from the institution.

Hardware components include:

• ESP32 microcontroller for IoT communication

• Current sensor for monitoring equipment power usage

• Temperature sensor for environmental monitoring

• PIR motion sensor for activity detection

• Camera module for visual monitoring

• Moisture sensor for environmental condition tracking

• AQI sensor (optional) for air quality monitoring

The microcontroller collects sensor data and transmits it to the central server through WiFi using IoT protocols.

3. Communication and IoT Layer

The communication layer enables data transfer between embedded devices and the central server.

Technologies used include:

• MQTT protocol for lightweight IoT communication

• HTTP / REST APIs for system integration

• WebSockets for real-time monitoring dashboards

This layer ensures reliable and continuous transfer of environmental and infrastructure data from institutional facilities.

4. OCR and Document Processing

Accreditation systems rely heavily on institutional documents such as SSR reports, IIQA data, research publications,

and accreditation submissions. Many of these documents exist in scanned PDF format.

Optical Character Recognition (OCR) is used to extract text from these documents.

Recommended OCR tools:

• Tesseract OCR

• AWS Textract

• Google Vision OCR

• PaddleOCR

OCR converts scanned documents into machine-readable text that can be analyzed by the AI system.

5. Large Language Model (LLM) Integration

EduTrace-ES integrates a Large Language Model (LLM) to analyze institutional documentation and generate insights.

Possible LLM choices include:

• GPT models

• Llama 3

• Mistral

The LLM analyzes extracted text from accreditation documents to identify gaps, measure institutional performance indicators,

and simulate accreditation readiness scoring.

6. Retrieval-Augmented Generation (RAG)

To effectively process large amounts of institutional data, the system uses Retrieval-Augmented Generation (RAG).

In this approach:

1. Documents are converted into embeddings.

2. Embeddings are stored in a vector database.

3. The LLM retrieves relevant information from the database before generating responses.

Vector databases that can be used include:

• FAISS

• Pinecone

• Weaviate

7. Backend System

The backend system manages data processing, API services, and AI workflows.

Recommended technologies:

• Python programming language

• FastAPI for high-performance API development

• Data pipelines for sensor ingestion and document analysis

The backend connects embedded devices, OCR systems, AI models, and the frontend dashboard.

8. Database Layer

Different databases are used depending on the type of data.

Structured Data:

• PostgreSQL

Sensor and time-series data:

• InfluxDB

Document and metadata storage:

• MongoDB

This hybrid data architecture ensures efficient storage and fast retrieval.

9. Frontend Dashboard

A web-based dashboard allows administrators to monitor institutional performance.

Frontend technologies:

• React

• Next.js

The dashboard displays:

• Accreditation readiness score

• Sensor monitoring data

• Environmental analytics

• AI-generated improvement suggestions

• Institutional performance indicators

10. Deployment and Infrastructure

The system can be deployed on cloud platforms or institutional servers.

Possible platforms include:

• Amazon Web Services (AWS)

• Google Cloud Platform

• Microsoft Azure

Containerization tools such as Docker can be used to simplify deployment and scaling.

11. Conclusion

The EduTrace-ES technology stack integrates IoT sensors, OCR-based document extraction, Large Language Models,

vector databases, and modern web technologies to create a comprehensive accreditation readiness monitoring system.

This architecture allows institutions to continuously evaluate their infrastructure, documentation quality,

research output, and institutional performance indicators while receiving AI-driven insights for improvement.